BUSINESS CASE: DATA MINING FOR HEART DISEASE PATTERN DISCOVERY AND RISK PREDICTION

1. BUSINESS CASE

Healthcare organizations sit on vast amounts of unutilized historical patient data. This project proposes a comprehensive Data Mining initiative to extract hidden patterns, identify complex risk factors, and build predictive models for heart disease based on standard clinical metrics. By systematically mining our Electronic Health Records (EHR), the organization can discover non-obvious patient risk archetypes, optimize clinical resource allocation, and proactively manage cardiovascular health, shifting our approach from reactive treatment to data-driven prevention.

2. PROBLEM STATEMENT

Currently, identifying patients at high risk for heart disease relies heavily on isolated test results and generalized medical guidelines.

- **Analytical Challenge:** The organization possesses rich data on patient demographics and vitals, but lacks the tools to systematically mine this data for complex, multi-variable risk patterns.
- **Clinical Challenge:** Non-traditional or subtle combinations of symptoms (e.g., specific blood pressure trends combined with age and asymptomatic chest pain) are easily missed by human review until a severe cardiac event occurs.
- **Operational & Financial Challenge:** Inefficient patient triage leads to workflow bottlenecks for cardiologists and exponentially higher costs associated with late-stage emergency interventions.

3. PROPOSED APPROACH

The application of standard data mining methodologies to our clinical datasets. The solution will utilize multiple data mining techniques:

- **Predictive Data Mining (Classification):** Algorithms (like Decision Trees or Naive Bayes) to assign a probability score to new patients, indicating their likelihood of having heart disease based on historical mined data.
- **Descriptive Data Mining (Association):** Techniques to segment patients into distinct "risk archetypes" and discover association rules (e.g., "If patient is Male, >50 yrs old, and has Flat ST Slope, then 85% probability of heart disease").

4. IMPLEMENTATION PLAN (YOUR TASKS)

PHASE 1: BUSINESS & DATA UNDERSTANDING

- Perform exploratory data analysis (EDA) to understand distributions and initial correlations within the 12 attributes.
    - Descriptive Statistics
    - Visualizations through charts and graphs

PHASE 2: DATA PREPARATION & MODELING

- Apply data mining algorithms to the historical data. Extract association rules to understand symptom combinations and build predictive classification models.
- Perform Data Cleaning and Preparation on the heart.csv dataset. (e.g. One-Hot Encoding on necessary features)
- Perform Association Rule Mining
    - Extract association rules to understand symptom combinations to build predictive classification models.
- Train two classification models using a decision tree and an algorithm of your choice
    - Evaluate and compare the performance of the two trained models by using the appropriate evaluation metrics.

PHASE 3: EVALUATION & INSIGHT GENERATION

- Present the discovered patterns and model accuracy. Ensure the mined insights make clinical sense (mitigating spurious correlations).
- Present and discuss mined insights and propose appropriate strategies or solutions through initiatives that can help address the problem statement.

GENERAL INSTRUCTIONS:

Your group will assume the role of data analysts working for a large hospital network. Your job is to bridge the gap between raw clinical data and strategic hospital operations. You have been tasked with finding a data-driven way to improve the early detection of heart disease.

Write a case study report for the Hospital Executive Board (e.g., the Chief Medical Officer and the Hospital Administrator). The report should be a professional document that details the data mining process used to address heart disease risk prediction. It must follow the phases outlined in the implementation plan. There should be sufficient discussions explaining the data mining process performed and finding acquired. Submit the Python notebook used as reference for your outputs.

A submission link will be provided for your outputs follow the file naming conventions below:

- Case Study Report: Lastnames_CCS230_Finals.pdf
- Python notebook: Lastnames_CCS230_Finals.ipynb

Submission deadline: May 18, 2026 | 11:59 PM (Tentative)

Example Report Outline:

1. Executive Summary (1 Paragraph)

- Briefly state the problem (identifying high-risk heart disease patients early).
- Summarize your primary recommendation (e.g., "We recommend deploying the Decision Tree model for initial triage...").
- Highlight the main expected business/clinical value (e.g., reduced mortality, optimized cardiologist scheduling).

2. Methodology Overview (Brief)

- Describe the dataset in plain English (e.g., "We analyzed historical data from 918 patients, looking at 11 key clinical and demographic metrics.").
- Briefly mention the techniques used (Rule Mining, Classification, Clustering) without getting bogged down in the math.

3. Key Clinical Insights (The "What")

This is where you present the patterns you discovered in the data.

- Association Rules: Highlight 2-3 of the interesting rules you found. Translate them into plain English (e.g., *"We found with 85% confidence that older males with asymptomatic chest pain have a high risk of heart disease."*).

- Patient Archetypes: Discuss the clusters you found using K-Means. Describe the "profiles" of the patients and how a hospital might use these profiles to tailor preventative care.

4. Model Evaluation & Selection (The "Why")

Compare your trained models.

- Performance: Which model was more accurate overall?
- The Clinical Context: Discuss your Confusion Matrix. Explain to the board why Recall (minimizing False Negatives) is critical in this specific medical context. What is the danger of a False Negative vs. a False Positive?
- Trade-offs: Discuss the trade-off between the *interpretability* of the Decision Tree (which doctors like) and the *predictive power* of the Random Forest.

5. Strategic Recommendation & Next Steps (The "How")

- Make a definitive choice. Tell the board exactly which model they should deploy in the Electronic Health Record (EHR) system and why.
- Suggest 1-2 potential risks of deploying this AI model and how the hospital can mitigate them (e.g., data bias, doctor pushback, or changing patient demographics over time).

Note: Do not rely on text alone. Include your Decision Tree plot, your Confusion Matrix, or your Feature Importance chart, but you must add captions explaining what the hospital board is looking at.

GRADING RUBRICS

Criterion	5 - Excellent	4 - Proficient	3 - Competent	2 - Developing	1 - Beginning
1. Exploratory Data Analysis & Preparation	Performs comprehensive EDA with insightful visualizations. Flawlessly applies necessary transformations (One-Hot Encoding for classification, logical discretization/binning for).	Performs solid EDA. Data is correctly encoded and binned, though visualizations or bin boundaries may lack deep justification.	Performs basic EDA. Data preparation has minor issues (e.g., dropping useful columns or sub-optimal bin sizes).	Missing either EDA or proper data preparation. Models run, but on poorly prepared or unscaled/un-encoded data.	No meaningful EDA performed. Data preparation is completely incorrect or missing, causing downstream algorithm errors.
2. Association Rule Mining	Perfectly executes association rule mining with well-chosen support/confidence thresholds. Extracts and deeply analyzes the top rules, explicitly translating them into actionable clinical insights.	Executes association rule mining correctly. Identifies relevant rules and provides a basic explanation of what the rules mean in a medical context.	Executes association rule mining but uses default or arbitrary thresholds resulting in too many or too few rules. Minimal interpretation of the output.	Attempts association rule mining but makes logical errors (e.g., failing to filter for rules where Heart Disease is the consequent).	Fails to implement the association rule mining algorithm or generates completely unreadable/irrelevant outputs.
3. Classification Modeling Implementation	Flawlessly implements both Decision Tree and chosen algorithm. Extracts highly readable rules from the tree and plots feature importance for the forest.	Implements both models correctly. Successfully plots the Decision Tree and extracts feature importance,	Implements both models, but struggles with the extraction of rules from the Decision Tree or feature importance from the	Only implements one of the required models, or makes a severe methodological error (e.g., testing on the	Fails to implement classification models or code does not compile/run.

		though visuals may lack polish.	Random Forest.	training dataset).	
4. Clinical Evaluation & Metrics (Context)	Displays profound understanding of the medical context. Explicitly prioritizes **Recall** (minimizing False Negatives) over raw Accuracy, supported by a thorough analysis of the Confusion Matrix.	Correctly evaluates models using Accuracy, Precision, and Recall. Mentions False Negatives/Positives but with less emphasis on the specific medical consequences.	Evaluates models but relies entirely on raw Accuracy. Includes a Confusion Matrix but does not explain its significance to the hospital.	Incomplete evaluation. Missing key metrics (Precision/Recall) or the Confusion Matrix.	No evaluation metrics provided, or completely misinterprets what the metrics mean.
5. Management Report & Synthesis	Highly professional report. Masterfully compares the distinct value of Association Rules vs. Predictive Probabilities. Makes a strongly justified, data-driven recommendation for hospital deployment.	Well-written report. Clearly explains the findings and makes a logical recommendation for deployment based on the data.	Basic report. Summarizes what was done but lacks a deeper comparison between the techniques. Recommendation is weak or arbitrary.	Report is poorly structured, confusing, or simply lists code outputs without translating them into business or medical value.	No management report provided, or text is entirely unrelated to the data mining findings.